Lean proof simplification after the word and controller constructions

A fourth review with all earlier proposals assumed available

Source commit `09d65a93c9df8684c1a226341d6a5120623d1cdb`

1 Findings and scope

The clearest next step is small, proof-local deletion, not another general framework. The new physical-order proof reconstructs finite-chain rigidity already available in Mathlib. The new auxiliary-cardinality proof reconstructs a standard divisibility fact. A three-client modular helper also costs more source than direct rewrites. These are specific new-client improvements after the earlier abstractions are assumed available; they are not new mathematical library results.

Together the three proof-only candidates remove 45 physical lines. A separate opportunity is semantic rather than compressive: derive the exact response of the constructed word on the twenty-field terminal band. The old source proves only the last two product fields. The stronger statement identifies every surviving marker term, including the exceptional bare contribution at field 2008. Its helper and compatibility costs must be paid explicitly.

Candidate	Physical source delta	Recommended treatment
Finite-order rigidity client	$180 \to 156$; -24	First proof-only trial
Coprime cardinality client	$50 \to 44$; -6	Independent proof cleanup
Three conditional H_s residues	$184 \to 169$; -15	Keep kernel symbolic
Finite predecessor representation	Additional -7	Optional definition edit
Marker-only terminal band	Positive API cost	Optional stronger theorem

Status of every new Lean candidate: uncompiled. No Lean, Lake or Elan executable is available in this task environment. No toolchain installation, new Lean build, kernel check or timing experiment was performed. Pinned-source inspection, mathematical arguments, patch checks and independent finite tests are useful evidence, but none substitutes for elaboration. The supplied PDF was built separately; that does not validate Lean code.

Pinned scope. The main proof tree contains 664 Lean files and 84,248 physical lines at the source commit. The comparison starts at `1e2a853935a6155cbcf2aa199aaca51946c4fff9`, the previous review cutoff. The final bounded exclusion check at `663fb267f6142ca4c0ea5bcd2d4f607f9a4b0eda` covers 28 direct manuscripts: 23 placed packets and five incoming archives. All four retained target files are unchanged at that later cutoff. The baseline is counterfactual: all useful prior contracts are available, even when deferred or represented by mutually incompatible alternatives. It is not a claim that all old implementations form one compiled tree.

No endpoint inflation. Existing native completeness and newer concrete word/controller construction progress must not be confused with a completed converse from every relevant arithmetic witness to the required construction. Reverse arithmetic and controller obligations remain open. Nothing here establishes the full polynomial–Con(ZFC) equivalence, changes its logical trust boundary, measures speedup or certifies a new kernel result.

2 What changed and what is already excluded

The new proof surface is concrete construction. From the previous cutoff to the source pin, 27 commits affect 36 active Lean files, with 4,107 inserted and 270 deleted lines. This is change volume, not simplification benefit. The source now constructs actual cyclic words and bounded carries (`d64aef2aa`); exact interval subsets (`e73271b71`); strict transport, dimensions and affine subsets (`41fc69695`); the scalar low mask (`5e8153809`); and every affine controller residue with one Boolean term (`367afdf46`). Copy masks, stronger word bounds and complete physical ordering follow in `33a8d7327`. The opportunities below live in this fresh surface; a new filename alone proves no novelty.

Upstream adoption is not credited again. At the source pin, the adoption ledger reports 77 lines removed by shared Pell estimates, 46 by signed-root recovery, 42 by internal natural-case reuse and 27 by pairing projections: 192 lines. At the final bounded cutoff it additionally records 638 lines across six modules from coefficient bounds, decoder invariants, evaluator/carrier wrappers and `Option.map_injective`. These are old proposals adopted upstream, not fourth-round savings; their recorded builds were not rerun. In particular, the coefficient file now adopts packet d's 127-line candidate, with a larger recorded object size. Source reduction is not evidence of speedup.

The source pin and final exclusion pin have different jobs. Candidate text and costs use `09d65a93c9df8684c1a226341d6a5120623d1cdb`. The final bounded review uses `663fb267f6142ca4c0ea5bcd2d4f607f9a4b0eda`. Eleven selected source/dependency files, including all four retained targets, are byte-identical between them. The frozen worktree was not moved.

The final corpus contains 28 manuscripts. At the source pin it had 27: 23 placed packets a–w and four incoming archives. The final check adds one incoming manuscript. Packets a–r have unchanged tree identities from the prior cutoff; s/t/u report sources match their formerly incoming archives. Our own previous delivery is also byte-identical and remains excluded. Movement is not novelty.

The inherited semantic screen covers a–u; fresh proposal bodies, qualifications and relevant notes cover v/w, the external incoming reports and the final additional manuscript. The latter includes helix period-one reuse, coprime divisibility, geometric identities, positive margins, exact signed-prefix borrow semantics and rank comparison through $C + qF$. All are assumed applied. None states the retained exact client deletions or the marker-only cyclic response. This is not a new line-by-line reading of every historical snapshot.

Important exclusions. Prior residue/support-zero interfaces, coherent word/carry views, finite-fiber sums, sparse placement, mask APIs, physical certificate assembly and rank/cardinality arguments remain available. Helix, canonical-band, shared-edge and exact-row proposals are excluded, as are our prior wall estimate, receiver causality and actual-row coherence.

The novelty test is marginal. What exact obligation remains after the nearest old contract is available? Count every helper and compatibility cost. A shorter displayed client is not a reduction if its new helper is larger. The final ledgers retain both the discarded projection idea and the reason it was removed.

3 Reuse rigidity of a finite ordinal

Target. WangTiles942PhysicalOrder.lean:68–97 proves that a strictly increasing self-map of $\mathrm{Fin}(n)$ fixes every point. The current proof obtains surjectivity, performs strong induction on the numerical index and splits the preimage into three cases. The exact conclusion follows from the existing rigidity theorem for finite order isomorphisms.

Mathematical proof. A strictly monotone map f is injective. Since domain and codomain are the same finite set, it is surjective. Strict monotonicity on a linear order gives $f(i) \le f(j) \iff i \le j$, so this bijection is an order isomorphism. An order isomorphism takes the strict lower segment of i bijectively onto the strict lower segment of $f(i)$. These segments have respectively i and $f(i)$ elements, hence their values are equal. Extensionality in $\mathrm{Fin}(n)$ finishes. The empty case is vacuous and no positivity hypothesis on n is needed.

Exact replacement sketch. All order-isomorphism packaging is included:

```
theorem strictMono_fin_eq_self {n : ℕ} (f : Fin n → Fin n)
    (hf : StrictMono f) (i : Fin n) : f i = i := by
  let e : Fin n ≃o Fin n :=
    ⟨Equiv.ofBijective f
      ⟨hf.injective, Finite.surjective_of_injective hf.injective⟩,
      hf.le_iff_le⟩
  exact Fin.ext (Fin.coe_orderIso_apply e i)
```

The shipped candidate uses the same proof with the original declaration header; its declaration is 30 to 6 physical lines and its body 28 to 4. The whole file is 180 to 156. The wrapping above is for page readability, not the measurement. No helper, import or downstream edit is added. In particular, `rankMap_eq`, physical positions and the Start convention are unchanged.

Pinned library support. Mathlib Order/Fin/Basic.lean:394–399 contains `Fin.coe_orderIso_apply`. Its module is already supplied by the physical-data import. This is exact client reuse, not a proposed new general rigidity lemma. The nearest older ideas are packet i's finite invariant-set surjectivity, packet k's exploratory order embeddings and packet s's prefix-rank/cardinality reasoning. None removes this new explicit induction.

Optional definition edit. At source lines 41–52, use the standard predecessor:

```
def nonStartIndex (j : Fin 941) : Fin 941 :=
  (abstractIndex j).pred (abstractIndex_ne_zero j)

theorem nonStartIndex_succ (j : Fin 941) :
    (nonStartIndex j).succ = abstractIndex j :=
  Fin.succ_pred (abstractIndex j) (abstractIndex_ne_zero j)
```

This saves another seven physical lines, yielding 149 combined. It changes a definition body, so it is separately selectable. Pinned Lean 4.32.0 defines `Fin.pred` by subtracting one; the nonzero proof and protected Start remain. A prepared `rfl` regression compares it with the old constructor, but has not been compiled. Test the proof-only variant first; if both are adopted use the supplied combined patch, not two overlapping whole-file alternatives.

4 Use the direct divisibility criterion for cardinality

Target. ControllerAuxiliaryChoice.lean:26–48 constructs a Boolean choice and a positive cardinality coprime to every power of the auxiliary prime. Its mathematical argument is already correct and its witness already explicit. The residual duplication is a hand-built bridge from nonzero reduction modulo the prime to nondivisibility of the least natural representative.

Mathematical proof. Let p be prime with $p \mid C$, and let $\phi : \mathbb{Z}/C \to \mathbb{Z}/p$ be the reduction homomorphism. Suppose $\phi(b) \neq 0$. At least one of $\phi(t)$ and $\phi(t-b)$ is nonzero: otherwise subtracting gives $\phi(b) = 0$. The existing Boolean-choice theorem therefore supplies $e \in \{0, 1\}$ such that, for $r = t - eb$, $\phi(r) \neq 0$.

Set $a = r.\mathrm{val}$. The standard cast identity identifies $(a : \mathbb{Z}/p)$ with $\phi(r)$. By $(a : \mathbb{Z}/p) = 0 \iff p \mid a$, we obtain $p \nmid a$. If $a = 0$, then $p \mid a$, a contradiction; hence $0 < a$. The canonical representative also satisfies $a < C$, while its cast in $\mathbb{Z}/C$ is exactly r. Finally, primality gives $\gcd(a, p^n) = 1$ for every n, including $n = 0$. Every witness and endpoint in the old theorem is retained.

Exact replacement sketch. After the existing derivation of `hr`, the fact that $\phi(r) \neq 0$, replace the local cast equality and contradiction chain with:

```
have hnd : ¬ p ∣ r.val := by
  rw [← ZMod.natCast_eq_zero_iff, ZMod.natCast_val]
  exact hr
have hpos : 0 < r.val :=
  Nat.pos_of_ne_zero fun hz => hnd (hz ▸ dvd_zero p)
```

Keep the original final tuple, including `hp.coprime_pow_of_not_dvd hnd`. The public declaration drops from 22 to 16 lines, and the whole file from 50 to 44. No helper or import is added. This is six lines of specific new-client reuse, not a new modular-arithmetic abstraction or a solution of the remaining power-orbit and safe-position obligations.

Pinned support and elaboration risk. Mathlib `Data/ZMod/Basic.lean` supplies `natCast_val` at line 263 and `natCast_eq_zero_iff` at line 518. Their exact statements are retained in the API receipt. Lean still needs to confirm type inference and the definitional match between `cast` and the local `phi`. If that reduction needs an explicit `change` or `simpa`, include its cost rather than silently preserving the advertised six-line saving.

A nearby proposal was withdrawn, not repackaged. An initial screen suggested shortening `ControllerResidueLift.exists_coordinate` with `castHom_apply`, `cast_eq_val` and `val_natCast`. A closer baseline check found that exact chain in packet p, `lean/DiophantineConsistency/SupportDifference.lean:27`, lines 27–34, already proposed as `residue_val`. Under the assumed-applied rule this is old infrastructure, even at a fresh client. The five-line projection patch was removed from the retained artifacts and all totals.

The nearest broader precedents for the surviving cardinality edit are E17 residue interfaces and LE24 cast normalization. They establish the approach, but not this exact nondivisibility-to-positive-coprime-witness deletion. The report credits only that residual client cost. This withdrawal is why a final accounting pass must follow semantic comparison, rather than treating an initial keyword screen as a novelty proof.

5 Remove a helper whose three clients are cheaper inline

Target. `CurrentControllerData2011.lean:150–182` has a private helper `hs_mod_of` for three conditional residues of

$$H_s = 1 + BK + B^2.$$

The helper proves a generic nested-modulus identity; each client supplies all intermediate residue values and the final finite calculation. Here the abstraction does not hide a difficult proof or simplify the total source. Deleting it and applying the modular laws at the three clients saves 15 physical lines, with no new helper or import.

Mathematical proof. For any modulus m, reduction respects addition and multiplication. If $B \bmod m = b$, $K \bmod m = k$, and $B^2 \bmod m = s$, then

$$H_s \bmod m = \big((1 \bmod m + (bk) \bmod m) \bmod m + s\big) \bmod m.$$

No cancellation, primality or invertibility is used in this calculation. The concrete inputs are:

m	$B \bmod m$	Assumed $K \bmod m$	$B^2 \bmod m$	$H_s \bmod m$
47	37	26	6	29
2011	1283	1995	1091	674
120199	37435	114787	99283	36379

These are conditional results. The K column is a hypothesis of each public theorem, not a newly checked evaluation of the actual 942-row kernel.

Exact proof pattern. Keep the existing public theorem headers and use:

```
attribute [local irreducible] CurrentKernel2011.K B

theorem Hs_mod_47 (hK : CurrentKernel2011.K % 47 = 26) :
    (1+B*CurrentKernel2011.K+B^2) % 47 = 29 := by
  rw [Nat.add_mod (1+B*CurrentKernel2011.K), Nat.add_mod 1,
    Nat.mul_mod B CurrentKernel2011.K,
    B_mod_47, hK, B_square_mod_47] <;> decide
```

The 2011 and auxiliary clients use exactly the same operations with their corresponding residue lemmas and `hK`. The supplied full-file candidate includes all three replacements and the complete deletion of `hs_mod_of`: 184 to 169 physical lines, 153 to 139 nonblank lines. The one-line discrepancy between these deltas is ordinary whitespace, not concealed proof code.

The abstraction boundary remains essential. The local irreducibility guard is preserved. It prevents elementary numeral reduction from unfolding $B = 2^{94517}$ and the literal physical permutation behind K. The direct rewrite must eliminate those symbolic occurrences through their supplied residue facts before `decide` is expected to finish. No recommendation is made to replace the guard with more permissive reduction or a larger heartbeat budget.

This is a deliberately narrow counterexample to "always factor repeated code." A shared helper earns its cost only if its total interface and proof are cheaper or materially safer than its clients. Here all three clients already need exact residue inputs, and the general law is standard library rewriting. If further clients arise, recompute the total cost rather than treating this decision as a permanent ban on a modular-expression API.

Validation gate. Compile this change independently after the coprime-cardinality edit, inspect all three theorem statements and axiom output, and confirm the module does not evaluate K. The source delta is established; successful elaboration, runtime and memory behavior are not.

6 An optional exact marker response on the terminal band

New conclusion rather than net compression. The source `CurrentWordTopBounds2011.lean:66–151` proves the last two convolution fields. Let w be the actual constructed cyclic word, m_j its numeric marker at cell j, and p_e, ω_e the actual kernel entry position and weight. The proposed stronger result is, for $1991 \le h < 2011$,

$$(\text{folded} * w)(\text{base}(j) + h) = \mathbf{1}_{h=2008}\, m_j + \sum_{e:\text{Entry}} \mathbf{1}_{h+p_e=2009}\, 2\omega_e m_j.$$

Here the indicators select terms, not new Boolean hypotheses. The tile assignment is arbitrary; dummy and marker maps take arbitrary natural values. Neither marked matching, a controller premise nor bounded carry data is required. The modulus remains $N \cdot 2011$ with $N > 0$, including $N = 1$.

Proof of the source restriction. The constructed local tile digit has no support beyond field 985. The dummy is at 977 and the marker at 1004. Therefore for $986 \le x < 2011$,

$$w(\text{base}(j) + x) = \begin{cases} m_j & x = 1004, \\ 0 & x \ne 1004. \end{cases}$$

The reflected source for kernel entry e is $h - (1005 - p_e) = h - 1005 + p_e$. Since $0 \le p_e \le 985$, the terminal band puts it in $[986, 1990]$. It survives precisely when $h - 1005 + p_e = 1004$, equivalently $h + p_e = 2009$. Equality is established in the actual cyclic ring before reading the field; no unchecked linear-index assumption is used.

The bare terms cannot all be discarded. The bare displacement 18 reads fields 1973–1992, outside support. Displacement 2020 reads residues 1982–2001 after crossing the cell boundary, also outside support. Displacement 1004 reads fields 987–1006, meeting the marker exactly when $h = 2008$. That bare contribution must be added even if a reflected entry also survives there.

```
theorem convolution_marker_tail
    (a : ZMod N → Fin 942) (dummy marker : ZMod N → ℕ)
    (j : ZMod N) (h : ℕ) (hl : 1991 ≤ h) (hh : h < 2011) :
    convolution folded (CurrentWordConstruction2011.word a dummy marker)
      (base j+(h : ZMod (N*2011))) =
      (if h = 2008 then marker j else 0) +
        ∑ e : Entry, if h+position e = 2009
          then 2*weight e*marker j else 0 := by
  -- Expand convolution_eq; remove the two outer bare terms.
  -- Rewrite the remaining bare source by word_marker_tail.
  -- Use top_reflected_source, then Finset.sum_congr.
  -- Each reflected term is the marker test above.
```

This is the exact theorem interface and a proof roadmap, not a compilable proof body. The candidate contains the complete proof and preserves all sixteen old theorem signatures. At $h = 2009$, injectivity of position leaves only Start at position zero. At $h = 2010$, no nonnegative position survives; these recover the two existing endpoints.

Sharpness and price. Extending to $h = 1990$ is false: the position-zero term reads supported field 985. The actual Start cell with zero controls gives a nonzero counterexample. The complete candidate grows 259 to 278 physical lines. Its two endpoint clients save 26 code lines, but three new declarations cost 43; after all other edits the net is *plus* 11 code lines and 353 lexical tokens. Adopt only if the full band is a useful consumer-facing contract, not to advertise those client-only savings.

7 Full costs and the marginal novelty boundary

Physical source costs are whole-file costs. They include every new declaration, preserved wrapper and changed local context. Negative deltas mean deletion. Report formatting, review scripts, source snapshots and extra check files are not production Lean reductions; they remain visible as review artifacts.

Choice	Before	After	Delta
Physical order isomorphism only	180	156	−24
Coprime-cardinality proof	50	44	−6
Three H_s proofs and helper deletion	184	169	−15
Proof-only core across three files	414	369	−45
Core with optional finite predecessor	414	362	−52
Core with optional marker-tail API	673	647	−26
All retained choices across four files	673	640	−33

The full word candidate adds 19 physical lines and 16 nonblank lines; after comments are removed, it adds 11 code lines. These are different measures, not conflicting totals. Its 353 extra lexical tokens reinforce the same conclusion: the stronger API is not an achieved compression. The physical predecessor and order-isomorphism alternatives overlap one file; do not add their standalone file sizes. The supplied combined variant is 149 lines.

Nearest prior contracts and what remains new.

- **Finite-order rigidity.** Packet i `report.tex`:222–249 uses finite surjectivity; packet k `novelty-audit.md`:67–69 explores order embeddings; packet s `evidence/novelty-audit-initial.txt`:254–260 discusses prefix ranks. Credit only removal of this new finite-ordinal induction, not those existing techniques.
- **Finite predecessor.** Standard core predecessor cancellation is old mathematics. No earlier exact `nonStartIndex` replacement was found. The separate definition-level edit is eligible only with its value-preservation check and downstream regression cost disclosed.
- **Cardinality and H_s.** E17/LE24 cover residue and cast techniques; E31 already projects stronger congruences to weaker moduli. The new deletions are respectively the exact prime-power-cardinality bridge and the demonstrably costly private helper at three new clients. The previously proposed `residue_val` projection is withdrawn entirely.
- **Terminal marker response.** E17 already supplies aligned cyclic sources and support-zero facts; E54 provides finite-fiber weighted sums, LE21 injective sparse placement, and E20 mask-block interfaces. The new conclusion is the exact twenty-field response, with its field-2008 collision, not generic residue or sparse-sum infrastructure. The incoming further-deltas P1 cutoff concerns a different coefficient-bound client.

Rejected extrapolations. The 59-chunk physical certificate assembly is covered by prior generic dispatch and is not a fresh suggestion. No extra abstraction is recommended for the short power-dimension construction. The marker-tail formula does not prove the full high mask, supply safe dummy positions, discharge actual kernel residues or establish a global equation converse. The old top carry and whole-word bounds already exist at this pin; preserving them is regression work, not a new result.

Interpretation. The measured deletion is modest but genuinely incremental. It is defensible precisely because the report removes the overlapping five-line projection idea, counts the marker API's growth, and distinguishes optional data-definition changes from proof-only replacements. No line count implies smaller proof terms, lower elaboration time or reduced kernel trust.

8 What is checked and how to adopt sequentially

Completed evidence. Candidate diffs replay exactly against the frozen source. Static checks preserve imports, public headers and relevant context commands; the only selected definition change is the optional `nonStartIndex`. The three stronger word declarations are explicit, and the sole removed private declaration is `hs_mod_of`. Scans find no new axiom, `sorry`, `admit` or `native_decide`. These scans do not inspect elaborated axiom dependencies.

Finite checks cover all 50,070 self-maps of sizes zero through six; the strict maps are identities. Physical data checks parse all 941 non-Start indices and 940 adjacent recorded-weight comparisons, but do not independently decode selector weights. Controller checks exercise finite coprimality cases and the three displayed scalar residues, not the unknown actual K residues. Word checks cover cyclic geometry, $N = 1$, numeric controls and counterexamples to the lower-threshold and omitted-bare-term mistakes. No finite test proves the universal Lean statements.

Required environment. Use an authorized complete checkout at the source pin, Lean `v4.32.0`, and the manifest-pinned Mathlib revision `81a5d257c8e410db227a6665ed08f64fea08e997`. Do not assume an existing successful `.olean` validates edited source. No dependency installation or Lean execution was performed here.

1. **Establish the baseline.** Verify the Git pin and source hashes. Build the unmodified target modules and existing check modules with the pinned dependencies. Record any pre-existing failure separately before attributing it to a patch.

2. **Try finite-order rigidity alone.** Apply `physical/patches/order_iso.patch`; rebuild `WangTiles942PhysicalOrder`, then its existing checks and the prepared regression. Check the order-isomorphism constructor and definitional forward map first if elaboration fails.

3. **Try the two controller edits separately.** Apply `controller_coprime_cardinality.patch`, rebuild `ControllerAuxiliaryChoice` and its checks, then its dependent residue-lift modules. Apply `controller_hs_inline.patch`; rebuild `CurrentControllerData2011` and its checks. Preserve irreducibility and all `hK` premises.

4. **Decide the predecessor change independently.** In a clean variant use `physical/patches/combined.patch` instead of the order-only patch. Compile the exact old-constructor `rfl` regression, the successor law, rank/position facts and existing consumers.

5. **Only then consider the stronger word API.** Apply `word/patches/marker_tail.patch`, rebuild `CurrentWordTopBounds2011`, its existing checks. Confirm the two endpoint wrappers, raw bounds, top carries and whole-word bounds still elaborate unchanged. Keep the bare 2008 term and test $N = 1$.

6. **Close integration rather than stopping at a leaf.** Rebuild affected downstream modules and the project umbrella. Inspect `#print axioms` for each changed/new theorem and compare with the project's policy. Record commands, exit codes, dependency revisions and source hashes. Only then label a variant Lean-validated.

Build sequentially: one process and one module only after its dependencies are ready. `LAKE_JOBS=1` does not enforce this; never use `LEAN_NUM_THREADS=0`. Rebuild changed imports before checks, and avoid umbrella fanout until the affected dependency chain is ready.

Stopping rule. Repair source/type-shape errors, refresh cost/hash receipts and rebuild. With Lean unavailable, leave the compilation gate explicitly open. If a positive-cost API has no actual downstream use, leave it optional rather than expanding scope to manufacture one.

9 Evidence map and research boundary

The archive separates proposals, receipts and inherited exclusions. The final artifact manifest is the authority for exact archive membership and hashes. It is not a claim that every file is Lean source or that all alternatives should be installed together.

Area	Purpose
`catalog/`	Initial 27-manuscript inventory and final 28-manuscript supplement, exact movement/prior-delivery comparisons, inherited semantic ledgers and the final exclusion/adoption decisions.
`evidence/`	Pinned source state, active Lean inventory and recent-commit boundaries; explicitly records absent compiler tools.
`physical/`	Order-only, predecessor-only and combined candidates; patches, pinned core/Mathlib receipts, regression and finite/static checks.
`controller/`	Retained coprime and H_s candidates/patches, complete size/hash manifest, API excerpts and guards. No withdrawn quotient-projection patch.
`word/`	Full positive-cost marker-tail candidate, preserved-wrapper patch, per-declaration accounting and finite-model controls.
`audit/`	Independent exact patch replay, static invariants and combined physical-line accounting. Its passing status is explicitly source/finite-only.
`report/`	Editable English LaTeX, rendered PDF and a build script using installed TeX files without downloading packages.

Read these receipts first. `catalog/EXCLUSION_AND_ADOPTION_LEDGER.md` records the source-pin decisions; `catalog/late663/FINAL_BOUNDED_REVIEW.md` completes the 28-manuscript comparison. The projection overlap is withdrawn. `catalog/movement_and_prior_delivery.json` distinguishes moved packets from new work. `audit/integrated_results.json` records exact patch composition and totals for the retained choices. The local module manifests and `word/accounting.json` provide per-file or per-declaration costs; their different nonblank/code metrics are labelled explicitly.

Source links are immutable. Project-source links use the source pin, not a moving main branch; the final exclusion/adoption supplement is explicitly pinned separately. Core API evidence uses Lean `v4.32.0`; Mathlib API evidence uses the exact revision above. A later source edit with the same theorem name can invalidate a patch or alter reduction behavior; compare bytes and rebuild rather than treating names as provenance.

The native endpoint already exists. The pinned equivalence plan, lines 629–646, records the completed marked-language/actual-blank-halting equivalence and `no_marked_word_iff_conZFC`. This is important positive progress, not an open native-completeness task. The same plan explicitly separates reverse arithmetic and controller construction needed for the actual equation equivalence. Its later construction checkpoints do not remove that distinction.

Remaining mathematical work is not hidden in these cleanups. Actual power-orbit identification, safe dummy placement, rank correction and remaining arithmetic witness obligations are not supplied by direct cast reuse or the terminal-band formula. The H_s wrappers still assume their actual K residues. A source-level equality of a new helper does not settle the project's internal-ZFC formalization or strengthen its stated axiom policy.

Recommendation. Validate the 45-line proof-only core first. Treat the seven-line predecessor edit as a separate representation-preservation decision. Retain the marker-tail theorem as a fully costed optional API until a real consumer needs its stronger interval conclusion. This leaves a small, auditable adoption sequence rather than another unverified umbrella refactor.